SYSC 5108 W26 Assignment 1

1. Imagine we have two possibilities to send a paper document: We can scan it and email the image, or we can use an optical character reader (OCR) to read the document and send the text file. Discuss the advantages and disadvantages of the two approaches in a comparative manner. When would one be preferable over the other? What is the implication for machine learning?

2. The following table lists a dataset of five individuals described via a set of stroke risk factors and their probability of suffering a stroke in the next five years. This dataset has been prepared by an analytics team who are developing a model as a decision support tool for doctors. The goal of the model is to classify individuals into groups on the basis of their risk of suffering a stroke STROKE RISK. In this dataset there are three categories of risk: low, medium, and high. All the descriptive features are Boolean, taking two levels: true or false.

ID	HIGH BLOOD PRESSURE	SMOKER	DIABETES	HEART DISEASE	STROKE RISK
1	true	false	true	true	high
2	true	true	true	true	high
3	true	false	false	true	medium
4	false	false	false	false	low
5	true	true	true	false	high

   a) How many possible models exist for the scenario described by the features in this dataset? A model should predict at least one outcome differently from other models.
   b) How many of these potential models would be consistent with this sample of data?

3. The table below shows a sample of a larger dataset containing details of policy holders at an insurance company. The descriptive features included in the table describe each policy holders' ID, occupation, gender, age, the value of their car, the type of insurance policy they hold, and their preferred contact channel.

ID	OCCUPATION	GENDER	AGE	MOTOR VALUE	POLICY TYPE	PREF CHANNEL
1	lab tech	female	43	42,632	planC	sms
2	farmhand	female	57	22,096	planA	phone
3	biophysicist	male	21	27,221	planA	phone
4	sheriff	female	47	21,460	planB	phone
5	painter	male	55	13,976	planC	phone
6	manager	male	19	4,866	planA	email
7	geologist	male	51	12,759	planC	phone
8	messenger	male	49	15,672	planB	phone
9	nurse	female	18	16,399	planC	sms
10	fire inspector	male	47	14,767	planC	email

   a) State whether each descriptive feature contains numeric, interval, ordinal, categorical, binary, or textual data.
   b) How many levels does each categorical, binary, or ordinal feature have?

4. What is the implication of the use of a single ϵ for all x_j in gradient descent?

5. Given the following dataset. Use the third sample point as the test data and the remaining as training data.

x	0.1	1.3	3.5	6.4	8.9
y	-1.2	2.7	1.4	4.3	2.6

   a) Find the optimal linear regression model manually by following slides 4 and 5 of Chapter 2. Show the squared test error. Draw your optimal function and dataset in a diagram.

b) Given $\beta = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, find the optimal learning rate ϵ^* and estimate the optimal parameter $\hat{\beta}$ using Newton's method. Discuss the difference between this one and the one in (a).
c) Find the optimal quadratic regression model. Show the squared error. Draw your optimal function and dataset in diagram.
d) Compare the errors in the above two models and discuss the impact of capacity.
e) Assume linear regression model and an initialized model parameter $\beta = (0.80 \quad 0.60)^T$. Suppose learning rate $\epsilon = 0.001$. Calculate one step update for the parameter β using backward error propagation as in slide 10 of chapter 2 and summed square error.

6. Suppose that x is a Boolean input with values -1 or 1. Design a perceptron that can predict the outcomes of NOT logic operation of its input.